

James Bray

R&D CONTRACTOR

Aerospace MEng graduate specialising in UAV system software, with a broad base of projects across firmware, controls and simulation. I drop into cross-disciplinary teams, move fast between system design and hands-on build, and ship working results.

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Experience

- Mission Software Engineer — [Windracers](#)

JUL 2022 - OCT 2022

 - Integrated features into the Mission Software using the C#/F.NET with Angular
 - Built custom STM32 development environments, migrating the team off outdated compilers
 - Operationalised a ML health monitor to detect and diagnose engine or sensor faults
 - Deployed custom AI microservices on AWS, including an automated ATC chatbot
 - Completed simulator and flight training on RC planes used for comms equipment testing
- Integration Engineer — [XPRIZE AURA](#)

SEP 2025 - JUN 2026

 - Deployed in Alaska to develop a swarm of autonomous UAVs for wildfire detection and mitigation
 - Engineered a novel thermal detection refinement method to localise fires with 1 m accuracy
 - Profiled the system architecture to identify and mitigate critical network bottlenecks
 - Ran rapid hardware-in-the-loop testing cycles across a 38 h non-stop development sprint
- Propulsion Avionics Engineer — [Sunride](#)

SEP 2024 - MAY 2026

 - Designed avionic systems for the university's first liquid rocket engine test stand
 - Used LabJack DAQ systems to interface sensors and actuators for precise control
 - Built a Simulink fluid model with PID control to regulate fuel and oxidiser flow
 - Implemented fault detection and safety protocols for engine test procedures
- Assistant Project Engineer — [AMRC](#)

JUL 2024 - OCT 2024

 - Designed and manufactured a £4,000 partial-discharge testing cell for 20 kV motor coils
 - Coordinated with 6 suppliers to source components for manufacturing
 - Managed deliverables, mitigated risks and produced supporting documentation
 - Used Ansys FEA to evaluate geometries for slinky-stator manufacture and a robot end effector

Education

- Aerospace Engineering MEng — [University of Sheffield](#)

SEP 2022 - MAY 2026

Masters, first-class. Avionics-focused: electronic systems and control theory applied to aerospace.

 - Dissertation: 'Risk-Aware Multi-UAV Search Strategies for Large-Scale Wildfire Detection' (86%)
 - Favourite modules: State-Space Control Design (92%), Multisensor Decision Systems (87%), Aircraft Dynamics & Control (85%), Real-Time Embedded Systems (79%), Data Modelling and Machine Intelligence (76%)
- A-Levels & GCSEs — [George Spencer Academy](#)

SEP 2015 - JUL 2022

 - A-Levels: A*AAA in Maths, Further Maths, Computer Science, Physics
 - GCSEs: 4 Grade 9s (Maths, Biology, Chemistry, Physics); grade 7–8 in Computer Science, D&T, Geography & English

Proficiencies

TECHNICAL	SOFTWARE	STRENGTHS
- Avionics & control systems - Algorithm development - Embedded / real-time firmware - Advanced manufacturing	- Python, C, C# - MATLAB & Simulink - Git & version control - Unity, Blender	- Cross-disciplinary R&D - System optimisation - Analytical problem-solving - Leadership & collaboration